FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2021

TEST DATE: 17-02-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. $lgG = (I - Ig)S$

$10 \times G = (50 - 10)12$

$G = 48\ \Omega$

2. A

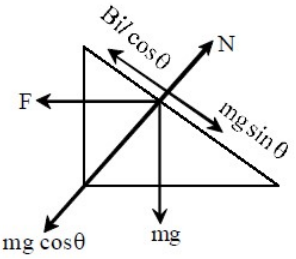
Sol. $\frac{hc}{\lambda} = a^2(z-1)^2 \Rightarrow \frac{\lambda}{\lambda_1} = \left(\frac{z_1 - 1}{z_1 - 1}\right)^2 = 4$

And $\frac{\lambda}{\lambda_2} = \left(\frac{z_2 - 1}{z - 1}\right)^2 = \frac{1}{4} \Rightarrow z_1 = 2z - 1$

$\Rightarrow \quad z_2 = \frac{z+1}{2}$

3. C

Sol.



$F\cos\theta = Mg\sin\theta$

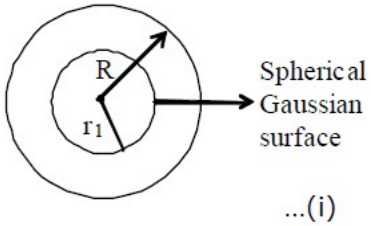
$B = \frac{mg\tan\theta}{i\ell}$

4. A

Sol. $\frac{I_E}{I_C} = 1.25 \Rightarrow \alpha = \frac{I_C}{I_E} = \frac{1}{1.25}$

5. A

Sol. $\oint \vec{E}.\vec{ds} = \frac{q_{in}}{\varepsilon_0}$



$E.4\pi r_1^2 = \frac{q_{in}}{\varepsilon_0}$

$E = \frac{q_{in}}{4\pi\varepsilon_0 r_1^2}$ ...(i)

Where $q_{in} = \int_0^{r_1} \rho dV = \int_0^{r_1} \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr$

$= 4\pi\rho_0 \left(\frac{r_1^3}{3} - \frac{r_1^4}{4R}\right)$

$\therefore$ From (i) $\boxed{E = \frac{\rho_0 r_1}{3\varepsilon_0}\left(1 - \frac{3r_1}{4R}\right)}$

6. B

Sol. $v_e = \sqrt{\frac{2GM}{R}} = 11.2 \text{ km/s}$

$v_e = \sqrt{\frac{2GM}{R/4}} = 2\sqrt{\frac{2GM}{R}} = 2 \times 11.2 = 22.4 \text{ km/s.}$

7. D

Sol. Image distance is 30 cm

$\frac{1}{F} = \frac{1}{f} + \frac{1}{f}$

$\frac{1}{30} = \frac{2}{f}$

As optical axis shifts down by (y – x) for second lens

$m = \frac{y - x - 1}{y - x}$

$y = 4.5$ cm

8. A

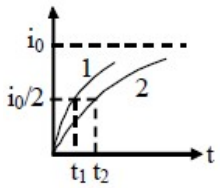
Sol. $u_1 = -u, v_1 = -\left(\frac{uf}{u - f}\right)$

$u_2 = \infty, v_2 = -f$

$|v_2| - |v_1| = \frac{uf}{u - f} - f = \frac{uf - uf + f^2}{u - f} = \frac{f^2}{u - f}$

9. C

Sol. Time taken for current to grow upto 50% is



$$t_{50\%} = \frac{L_1}{R}$$

$$t_{50\%} = \frac{L_2}{R}$$

$\therefore \quad t_2 > t_1$ (because $L_2 > L_1$)

Hence correct graph is 3

10. B

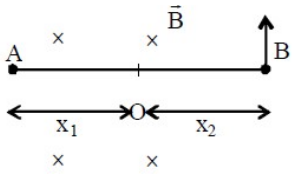
Sol. Ferromagnetic and not diamagnetic or paramagnetic

11. C

Sol. $y = \frac{2\lambda D}{d}$ is linear position from centre.

12. B

Sol.



$$\frac{B\omega x_2^2}{2} - \frac{B\omega x_1^2}{2}$$

$$\frac{B\omega}{2}\left(x_2^2 - x_1^2\right)$$

$$\frac{B\omega(x_2 + x_1)(x_2 - x_1)}{2}$$

$$\frac{B\omega\ell}{2}\left(x_2 - x_1\right)$$

$$\frac{B\omega\ell^2}{6} = \frac{B\omega\ell}{2}\left(\ell - 2x_1\right)$$

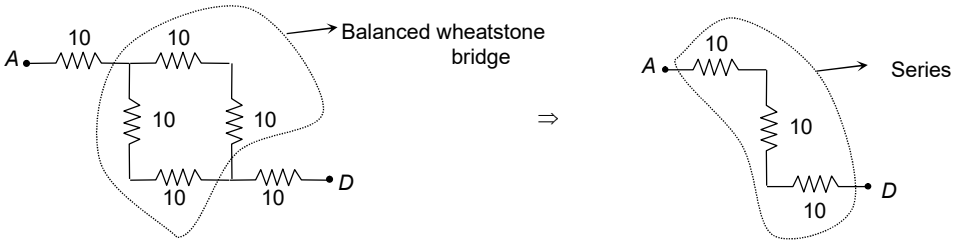
$$\frac{\ell}{3} = \ell - 2x_1$$

$$2x_1 = \ell - \frac{\ell}{3} = \frac{2\ell}{3}$$

$$x_1 = \frac{\ell}{3}$$

13. C

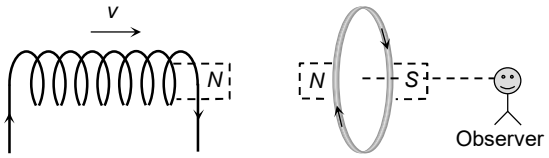
Sol. The equivalent circuit of above fig between *A* and *D* can be drawn as



So $R_{eq} = 10 + 10 + 10 = 30\Omega$

14. B

Sol. The direction of current in the solenoid is anti-clockwise as seen by observer. On displacing it towards the loop a current in the loop will be induced in a direction so as to oppose the approach of solenoid. Therefore the direction of induced current as observed by the observer will be clockwise.



15. A

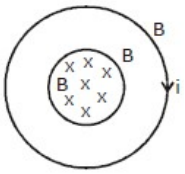
Sol. $i = 2t$

$\therefore$ i will increase with time

$\therefore$ $B\otimes\uparrow$

$\therefore$ $\phi\uparrow$

$\therefore$ Induced current will be ACW to produce magnetic field $\odot$ to oppose increasing flux (Lenz's law)

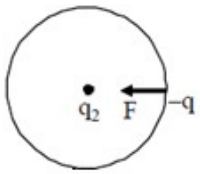


16. B

Sol. If balance length is not obtained then it means that potential difference across external resistance is more than potential difference across potentiometer wire.

17. A

Sol.



$F = F_C$

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{q_1q_2}{4\pi\varepsilon_0 rm}}$$

$$T = \frac{2\pi r}{v} = 2\pi r\sqrt{\frac{4\pi\varepsilon_0 m}{q_1q_2}}$$

$$T = \sqrt{\frac{16\pi^3 m\varepsilon_0 r^3}{q_1q_2}}$$

18. B

Sol. If the sheet is heated then both d_1 and d_2 will increase since the thermal expansion of isotropic solid is similar to true photographic enlargement.

19. C

Sol. Heat lost by *A* = Heat gained by *B*

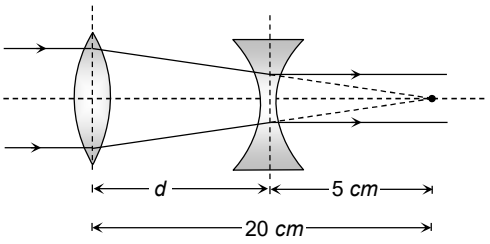
$\Rightarrow m_A \times c_A \times (T_A - T) = m_B \times c_B \times (T - T_B)$ Since $m_A = m_B$ and Temperature of the mixture (*T*) = 28°*C*

$\therefore c_A \times (32 - 28) = c_B \times (28 - 24) \Rightarrow \frac{c_A}{c_B} = 1:1$

20. B

Sol. From figure it is clear that separation between lenses

$d = 20 - 5 = 15\,cm$



SECTION – B

21. 00009.00

Sol. The final surface energy must be less than the original surface energy

$$4\pi\left(\frac{3r^2}{4}\right)^{2/3}\frac{TL}{\ell^{1/3}} < 2\pi rLT$$

$$\frac{9}{2}r < \ell$$

22. 00005.00

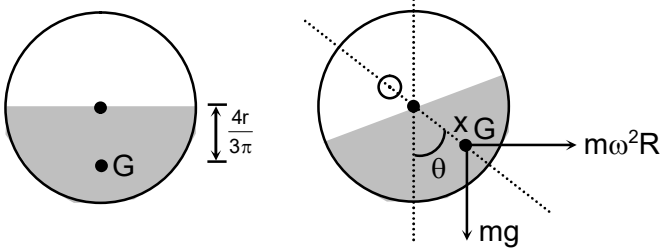
Sol. $I\alpha = -mgx\sin\theta$

$$\Rightarrow \frac{1}{2}mr^2\frac{d^2\theta}{dt^2} = -mgx\sin\theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{2g}{r^2}\left(\frac{4r}{3\pi}\right)\theta$$

$$\therefore \omega = \sqrt{\frac{8g}{3\pi r}} \Rightarrow f$$

= 0.46 Hz



23. 00000.15

Sol. Energy of incident photons in eV is given as

$$E = \frac{12431}{1800}eV$$

As work function of metal is 2eV, the maximum kinetic energy of ejected electrons is

$$KE_{max} = E - \phi$$

$$\Rightarrow \quad KE_{max} = 6.9 - 2eV$$

$$\Rightarrow \quad KE_{max} = 4.9eV$$

If v_{max} be the speed of fastest electrons then we have

$$\frac{1}{2}mv_{max}^2 = 4.9\times1.6\times10^{-19}\text{ joule}$$

$$\Rightarrow \quad v_{max} = \sqrt{\frac{2\times4.9\times1.6\times10^{-19}}{9.1\times10^{-31}}}$$

$$\Rightarrow \quad v_{max} = 1.31\times10^6\text{ m/s}$$

When an electron with this speed enters a uniform magnetic field normally it follows a circular path whose radius can be given by

$$r = \frac{mv}{qB} \qquad [\text{As } qvB = \frac{mv^2}{r}]$$

$$\Rightarrow \quad r = \frac{9.1\times10^{-31}\times1.31\times10^6}{1.6\times10^{-19}\times5\times10^{-5}}$$

$$\Rightarrow \quad r = 0.149\text{ m}$$

24. 00002.39

Sol. $E = C_1C_2\cos\omega_0 t + \frac{C_1C_3}{2}\cos(\omega_0+\omega)t + \frac{C_1C_3}{2}\cos(\omega_0-\omega)t$

Of the three components, the highest frequency component will liberate the electrons with maximum kinetic energy

$(K.E)_{\max} + \phi = \frac{h}{2\pi}(\omega_0+\omega) \Rightarrow \phi$ = 2.39 eV.

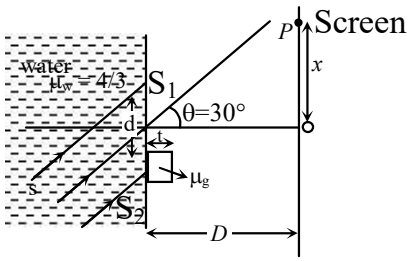
25. 00001.66

Sol. If central maxima is formed at point P on the screen, then

Path difference = $S_2P + (\mu_g - 1)t - S_1P - \mu_w d \sin 30° = 0$

$\Rightarrow \quad S_2P - S_1P = \frac{xd}{D} = \mu_w d\sin 30° - (\mu_g - 1)\times t$

$\Rightarrow \quad x = -1.66$ cm



26. 00007.60

Sol. Concept of vertical circular motion.

27. 00033.60

Sol. From the concept of charging of capacitor.

28. 00000.90

Sol. From the concept of work done by gas.

29. 00000.80

Sol. For water column in one arm of *U* tube, h_1 = 10.0 cm; ρ_1 (density = 1 gcm^{-3})

For spirit column in other arm of *U* tube, h_2 = 12.5 cm; ρ_2 = ?

As the mercury columns in the two arms of *U* tube are in level, therefore pressure exerted by each is equal.

Hence $h_1\rho_1 g = h_2\rho_2 g$ or $\rho_2 = \frac{h_1\rho_1}{h_2} = \frac{10\times 1}{12.5} = 0.8$ g cm^{-3}

Therefore, relative density of spirit = ρ_2/ρ_1 = 0.8 / 1 = 0.8

30. 00007.00

Sol. $(A_{CM})_x = \frac{mA + MA}{m+M} = A$

$(A_{CM})_y = \frac{M\times 0 + mR\alpha}{m+M} = \frac{mR\alpha}{m+M}$

$f = (M+m)A$ … (i)

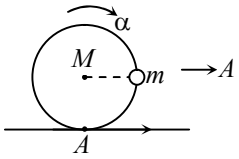
$(M+m)g - N = (M+m)(A_{CM})_y$ … (ii)

$mgR = I_A\alpha$ … (iii)

$A = R\alpha$ … (iv)

$\therefore$ N = 70 N

$\therefore$ n = 7



Chemistry

PART – II

SECTION – A

31. A

Sol. Δ corresponds to the wave length of the visible light and $\Delta = \frac{hc}{\lambda}$

The order of ability to produce d-orbital splitting of various ligands:

$CO > CN^- > NO_2^- > en > NH_3 > H_2O > OH^- > F^- > cl^- > Br^- > I^-$

strong field ligands (larger Δ) — weak field ligands (smaller Δ)

32. C

Sol. $\Delta T_b = \frac{1000 K_b w_1}{m_1 w_1}$

$m_1(A_2B) = 90 = 2A + B$

$2A + B = 90$

we get A = 40; B=10

33. C

Sol. pH changes from 0 to 7.

Accordingly $E_{red.}$ decreases by 0.059 $\log 10^{-7}$ i.e. 0.059 × (–7) = – 0.41 volt.

34. C

Sol. Factual.

35. C

Sol. Factual.

36. C

Sol. $SO_2 + Br_2 + 2H_2O \rightarrow 2HBr + H_2SO_4$

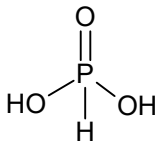
37. A

Sol. Eq. $KMnO_4$ = Eq. FeC_2O_4

5 × moles of $KMnO_4$ = 1 × 3

38. C

Sol.



$2 \times 20 \times 0.1 = 40 \times M$

$0.1 = M_{KOH}$

39. B

Sol. $K = \frac{0.693}{t_{50\%}} = \frac{0.693}{34.65} = 4 \times 10^{-2}$

40. D

Sol. Diamond has ZnS structure. In it r + r = $\frac{\sqrt{3}\times a}{4}$ (Nearest distance is between corner C atom and C atom in tetrahedral void).

41. C

Sol. Factual

42. D

Sol. $4KI + HgCl_2 \rightarrow K_2HgI_4 + 2KCl$

Remained KI = 5 – 4 = 1 mol

1 mol K_2HgI_4 and 2 mol KCl is formed.

$(K_2HgI_4 \rightarrow 2K^+ + HgI_4^{2-})$

$\left|(\Delta T_f)_o\right| = K_f(i_1m_1 + i_2m_2 + i_3m_3)$

$(T_f)_0 = -4.78^oC$

43. D

Sol. Because *P*ex = 0, -*P*exΔV = 0, so that no work is performed in this case.

44. C

Sol. All- cis –[10] annulene obeys (4n+2) π rule, but it is not aromatic because it is too strained when planar.

45. A

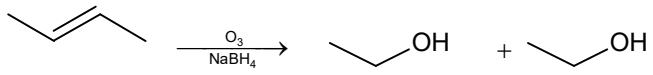
Sol. Small concentration of phenol (0.2%) acts as antiseptic while 1% solution of phenol acts as disinfectant.

46. A

Sol. acetaldehyde$(K_f = 1.06)$ acetone$(K_f = 0.001)$ Chloral$(K_f = 2000)$

Formaldehyde$(K_f = 2280)$ Hexafluoro acetone$(K_f = 1.2\times10^6)$

47. C

Sol. 

48. D

Sol. Factual

49. D

Sol. Factual

50. D

Sol. D-Fructose gives two products on reduction as a new chiral atom is formed at C–2 carbon.

SECTION – B

51. 00009.00

Sol. Basic Buffer

$$pOH = pK_b + \log_{10}\left(\frac{\text{Salt}}{\text{weak base}}\right)$$

$$pOH = 5 - \log_{10} 3 + \log_{10}(3)$$

$$pH = 14 - pOH$$

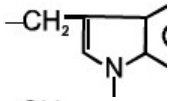
$$pH = 14 - 5 + \log_{10} 3 - \log_{10} 3$$

$$pH = 9$$

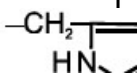
52. 00003.00

Sol. 1. Phenyl alanine - $-CH_2-C_6H_5$

2. Tryptophan -



3. Histidine -



53. 00113.60

Sol. First order type reaction follow, time= $4 \times t_{\frac{1}{2}} = 4 \times 28.4 = 113.60$

54. 00009.20

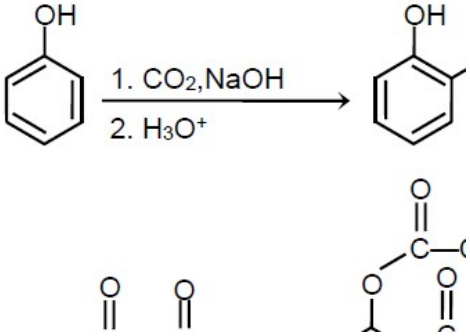
Sol. $\alpha = \frac{7.8}{390} = 2 \times 10^{-2}$

$K_a = c\alpha^2 = 16 \times 10^{-6}$

or $pK_a = 4.8$

55. 00018.00

Sol.



Molecular formula of Y = $C_9H_8O_4$

Molecular mass of (Y) = 108 + 8 + 64 = 180

56. 00000.25

Sol. $K_{eq.} = \frac{[B]}{[A]}$

$K_{1000} \frac{10}{1} = 10$ and $K_{2000} = \frac{100}{1} = 100$

Now, $\frac{\Delta G^{o}_{1000}}{\Delta G^{o}_{2000}} = \frac{(-RT\ln k_{eq})_{1000}}{(-RT\ln K_{eq})_{2000}} = \frac{1000\times\ln 10}{2000\times\ln 100}$

$= 0.25$

57. 00013.32

Sol. $E^{o}_{cell} = 1.23 - 0.00 = 1.23V$

$G^{o}_{cell} = nFE^{o}_{cell} = -2\times 96500\times 1.23J$

$\therefore$ Work derived from this fuel cell

$= \frac{70}{100}\times\left(-\Delta G^{o}_{cell}\right)\times 10^{-3} = xJ$

Since insulated vessel, hence q = 0

From equation, for monoatomic gas,

$w = \Delta U \Rightarrow x = nC_{V,m}\Delta T\left\{C_{V,m} = \frac{3R}{2}\right\}$

Or, $\frac{70}{100}\times\left(2\times 96500\times 1.23\right)\times 10^{-3} = 1\times\frac{3}{2}\times 8.314\times\Delta T$

$\therefore \quad \Delta T = 13.32$

58. 00006.15

Sol. $2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$

Moles of Al takes $= \frac{5.4}{27} = 0.2$

Moles of H_2SO_4 taken $= \frac{50\times 50}{1000} = 0.25$

As $\frac{0.2}{2} > \frac{0.25}{3}$, H_2SO_4 is limiting reagent

Now, moles of H_2 formed $= \frac{3}{3}\times 0.25 = 0.25$

$\therefore$ Volume of H_2 gas formed $= \frac{nRT}{P}$

$= \frac{0.25\times 0.082\times 300}{1} = 6.15L$

59. 00001.20

Sol. Initial moles of $U^{238} = \frac{68\times 10^{-6}}{238} = x$

Moles of U^{238} decayed in three half-lives $= \frac{7}{8}x$

In decay from U^{238} to Pb^{206}, each U^{238} atom decays and produces 8 α-particles and hence, total number of α-particles emitted out

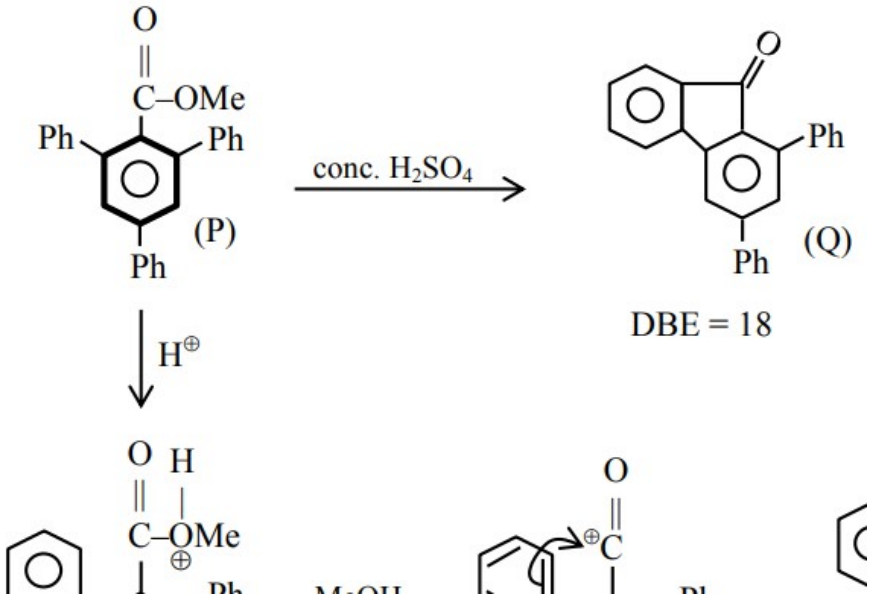
$= \left(\frac{7}{8}x\right)\times 8\times N_A$

$= 7\times\frac{68\times 10^{-6}}{238}\times 6.022\times 10^{23}$

$= 1.204\times 10^{18}$

60. 00018.00

Sol.



Mathematics

PART – III

SECTION – A

61. C

Sol. The asymptotes are $y = \pm \frac{b}{a} x$

$\Rightarrow \quad \tan 30° = \left|\frac{\frac{b}{a}+\frac{b}{a}}{1-\frac{b^2}{a^2}}\right| = \left|\frac{2ab}{a^2-b^2}\right| = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad a^2 - b^2 = \pm 2\sqrt{3}\, ab.$

Also $a^2 + b^2 = a^2e^2$

$\Rightarrow \quad e^4 - 16e^2 + 16 = 0$

$\Rightarrow \quad e^2 = 8 \pm 4\sqrt{3}$

$\Rightarrow \quad e = \sqrt{6} \pm \sqrt{2}.$

62. B

Sol. The pair of lines $x^2 - 3y^2 + 3xy + 3x(lx + my) = 0$ are perpendicular to each other.

$\Rightarrow 1 - 3 + 3l = 0 \Rightarrow l = \frac{2}{3}$

Hence the chords are $2x - 3 + 3my = 0$, which pass through the fixed point $\left(\frac{3}{2}, 0\right)$.

63. C

Sol. Major axis of hyperbola bisects the asymptote

$\Rightarrow$ equation of other asymptote $x = 2y$

equation of hyperbola $(y - 2x)(x - 2y) + k = 0$ it passes through (3, 4)

$\Rightarrow$ required equation $2x^2 + 2y^2 - 5xy + 10 = 0$.

64. A

Sol. Putting $x = \frac{3}{y}$ in $2x^2 + 2y^2 + 15x - 6y + 1 = 0$

We get

$2\left(\frac{3}{y}\right)^2 + 2y^2 + 5\left(\frac{3}{y}\right) - 6y + 1 = 0$

$\frac{18}{y^2} + 2y + \frac{15}{y} - 6y + 1 = 0$

$18 + 2y^4 + 15y - 6y^3 + y^2 = 0$

$2y^4 - 6y^3 + y^2 + 15y + 18 = 0$

$\sum yi = 3 \qquad \Pi yi = 9$

65. B

Sol. The equation of tangent is $y = mx \pm \sqrt{25m^2 - 16}$, it passes through (–2, 5)

$\therefore 5 = -2m \pm \sqrt{25m^2 - 16}$

$\Rightarrow (2m + 5)^2 = 25m^2 - 16$

$\Rightarrow 4m^2 + 20m + 25 = 25m^2 - 16$

$\Rightarrow 21m^2 - 20m - 41 = 0$

$\Rightarrow (m+1)(21m - 41) = 0$

$\Rightarrow [m_1] = -1, [m_2] = 1$

$\therefore$ equation of tangents are y = – x + 3 and y = x + 3 they intersect at (0, 3).

66. A

Sol. If $x_1x_2x_3 = 30$ then number of solution 3^3

If $x_1x_2x_3 = 15$ then number of solution 3^2

Similarly for $x_1x_2x_3$ equal to 10 & 6 solution are 3^2 & 3^2 and $x_1x_2x_3$ equal to 5, 3 & 2 solution are 3 each i.e. 3×3 and for $x_1x_2x_3$ = 1 there is just one solution

Total $= 3^3 + 3 \times 3^2 + 3 \times 2 + 1$ = 64

67. B

Sol. $f''(x) - 3f'(x) > 3$

$\Rightarrow \frac{d}{dx}\left(e^{-3x}f(x)\right) > 3e^{-3x} \Rightarrow \frac{d}{dx}\left(e^{-3x}f'(x) + e^{-3x}\right) > 0$

$\Rightarrow e^{-3x}\left(f'(x)+1\right)$ is increasing function

$e^{-3x}\left(f'(x)+1\right) > f'(0) + 1 \forall x > 0$

$\therefore f(x) + x$ is increasing function

68. C

Sol. $f'(x) = \cos\left(\frac{x^2 + 2x + 1}{5}\right) = \cos\left(\frac{x+1^2}{5}\right)$

Since $0 < x < 2, \frac{1}{5} < \frac{(x+1)^2}{5} < \frac{9}{5}$

$\cos\frac{(x+1)^2}{5} = 0$ only when $\frac{(x+1)^2}{5} = \frac{\pi}{2}$

$\therefore x = \sqrt{\frac{5\pi}{2}} - 1$ is a point of local maximum

69. B

Sol. $f(x) = \int_0^1 \frac{x^2 + t^2}{2 - t} dt$

$= \int_0^1 \left(\frac{x^2}{2-1} - t - 2 + \frac{4}{2-t}\right) dt$

$= \left(x^2 + 4\right) in2 - \frac{1}{2} - 2$

$\therefore y = x^2 ln2 + 4ln2 - \frac{5}{2}$

Which is a parabola

70. C

Sol. $\int_{-2}^{2}(y_2-y_1)dx=\int_{-2}^{2}\left(\frac{8}{x^2+4}-\frac{x^2}{4}\right)dx$

$=2\int_{0}^{2}\left(\frac{8}{x^2+4}-\frac{x^2}{4}\right)$

$=\frac{16}{4}\left[\tan^{-1}\frac{x}{2}\right]_0^2-\frac{1}{2}\left[\frac{x^3}{3}\right]_0^2$

$=\frac{8\pi}{4}-\frac{8}{6}$

$=2\left(\pi-\frac{2}{3}\right)$

71. C

Sol. $\frac{y^2}{2}-3xy+2y=x^2+x+c$

$y^2-6xy+4y=2x^2+2x+c$

$y^2-2x^2-6xy+4y-2x=c$

72. B

Sol. $\sum x=40\times200-50+40=7990$

Corrected $\bar{x}=\frac{7990}{200}=39.95$

Incorrect $\sum x^2=n\left[\sigma^2+\bar{x}^2\right]=200\left[15^2+40^2\right]=365000$

Correct $\sum x^2=36410-50^2+40^2=364100$

Correct $\sigma=\sqrt{\frac{364100}{200}-(39.95)^2}=14.98$

73. B

Sol. $\frac{dy}{dx}=\frac{x^2-4x+3}{x^2-7x+10}=\frac{(2x-4)(x^2-7x+10)-(2x-7)(x^2-4x+3)}{(x^2-7x+10)^2}$

$=2x^3-14x^2+20x-4x^2+28x-40$

$\frac{-2x^3-8x^2-6x+7x^2-28x+21}{(x^2-7x+10)^2}=\frac{-19x^2+14x-19}{(x^2-7x+10)^2}<0$

74. A

Sol. $(\vec{a}\times\vec{c}).\vec{a}=\vec{b}.\vec{a}+\lambda\vec{a}.\vec{a}$

$\lambda=-\frac{1}{3}$

Taking cross product $(\vec{a}\times\vec{c})\times\vec{a}=\vec{b}\times\vec{a}+0$

$\Rightarrow|\vec{a}|^2\vec{c}-(\vec{a}.\vec{c})\vec{a}=\vec{b}\times\vec{a}\Rightarrow\vec{c}\,\frac{(\vec{b}\times\vec{a})+(\vec{a}.\vec{c})\vec{a}}{|\vec{a}|^2}$

$(\vec{a}\times\vec{b}).\vec{c}=\frac{|\vec{b}\times\vec{a}|^2}{|\vec{a}|^2}=-\frac{8}{3}$

75. A

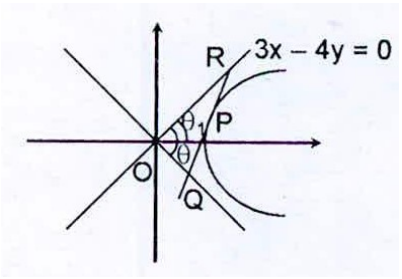
Sol. $\tan\frac{\theta}{2}=\frac{3}{4}$

$\Rightarrow \sin\theta=\frac{24}{25}$

$\Rightarrow$ Area of ΔORQ is

$\frac{1}{2}\times OR\times OQ\times\frac{24}{25}=3\times 4$

$OR\times OQ=25$



76. A

Sol. Let $\frac{y+x}{\sqrt{2}}=x$, $\frac{y-x}{\sqrt{2}}=y$

$\frac{x^2}{8}-\frac{y^2}{4}=1$

$a^2=8, b^2=4$

$e=\sqrt{\frac{12}{8}}=\sqrt{\frac{3}{2}}$

$x=\pm ae, y=0$

$\frac{y+x}{\sqrt{2}}=\pm\sqrt{8}\times\sqrt{\frac{3}{2}}, \frac{y-x}{\sqrt{2}}=0$

77. D

Sol. $\frac{{}^{10}C_7\times{}^6C_2}{9\times9\times8\times7\times6\times5\times4}=\frac{10\times9\times8\times6\times5}{3!x\times2!x\times9\times9\times8\times7\times6\times5\times4}=\frac{10}{6\times2\times9\times7\times4}=\frac{5}{1512}$

78. D

Sol. x = 2R cos A, y = 2R cos B, z = 2R cos C

$\frac{xy}{ab}=\frac{4R^2\cos A\cos B}{ab}=\cot A\cot B$

$\Rightarrow \frac{xy}{ab}+\frac{yz}{bc}+\frac{zx}{ac}=\cot A\cot B+\cot B\cot C+\cot C\cot A=1$

79. A

Sol. Let α is angle made by OP with x-axis and let θ be angle made by tangent at P with x-axis giving

$(\theta+2\alpha)=\frac{\pi}{2}$

$\tan\theta=\frac{dy}{dx}$ and $\tan\alpha=\frac{y}{x}$

80. B

Sol. Make truth table.

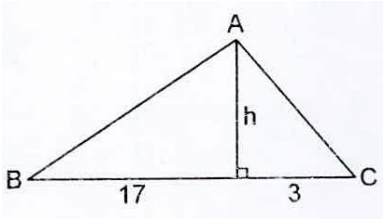
SECTION – B

81. 00005.00

Sol. $B + C = \pi - A$

$\tan(B + C) = -\tan A$

$$\frac{h\left(\frac{1}{17}+\frac{1}{3}\right)}{1-\frac{h^2}{51}} = -\frac{22}{7} \Rightarrow \left(\frac{20h}{51-h^2}\right) = -\frac{22}{7}$$



Area of triangle ABC $\Delta = -\frac{1}{2} \times 20 \times 11 = 110$

82. 00003.00

Sol. Area of square $2(a^2 + b^2) = 10$

$\Rightarrow a^2 + b^2 = 5, (a - b) = 1$

$\Rightarrow ab = 2$ and $a + b = 3$

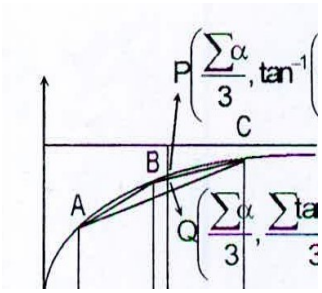
$\Rightarrow a = 2, b = 1, e = \sqrt{\frac{3}{4}} \quad \Rightarrow 4e^2 = 3$

83. 00005.00

Sol. At least one books can be selected in $(m + 1)^n - 1$

84. 00002.00

Sol. $$\frac{\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma}{3} \le \tan^{-1}\left(\frac{\alpha+\beta+\gamma}{3}\right) = \frac{\pi}{6}$$



$$\Rightarrow \frac{\pi}{\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma} \ge 2$$

85. 00001.00

Sol. Let $\tan\frac{\theta}{2} = x$

$$\Rightarrow \frac{1}{x^2} = \frac{1+x^2}{1-x^2} - \frac{1-x^2}{1+x^2} \quad \Rightarrow \frac{1}{x^2} = \frac{4.x^2}{1-x^2}$$

$\Rightarrow 1 - x^4 = 4x^4 \Rightarrow 5x^4 = 1$

$\Rightarrow 5\tan^4\frac{\theta}{2} = 1$

86. 00005.76

Sol. Point is $(a\sec\theta, b\tan\theta)$

asymptotes are $bx - ay = 0$ and $bx + ay = 0$

Now $\frac{ab\sec\theta - ab\tan\theta}{\sqrt{a^2+b^2}} \times \frac{ab\sec\theta + ab\tan\theta}{\sqrt{a^2+b^2}}$

$= \frac{a^2b^2}{a^2+b^2} = \frac{144}{25}$.

87. 00001.66

Sol. $b + c = 4a$

$\sin B + \sin C = 4\sin A$

$$\cos\frac{B-C}{2} = 4\cos\frac{B+C}{2}$$

$$\cos\frac{B}{2}\cos\frac{C}{2} + \sin\frac{B}{2}\sin\frac{C}{2} = 4\left[\cos\frac{B}{2}\cos\frac{C}{2} - \sin\frac{B}{2}\sin\frac{C}{2}\right]$$

$$3\cos\frac{B}{2}\cos\frac{C}{2} = 5\sin\frac{B}{2}\sin\frac{C}{2}$$

$$\cot\frac{B}{2}\cot\frac{C}{2} = \frac{5}{3}.$$

88. 03456.00

Sol. $y = ax^2 + bx + c$, vertex is (4, 2)

$\Rightarrow 4 = -\frac{b}{2a}$, $b = -8a$, $\frac{4ac - b^2}{4a} = 2 \Rightarrow c = 2 + \frac{b^2}{4a} = 2 + 16a$

Now $\Delta = abc = -8a^2(2 + 16a) = -16(a^2 + 8a^3)$

$\Rightarrow \frac{d\Delta}{da} = -16(2a + 24a^2) < 0 \quad \forall\ a \in [1, 3]$.

$\Rightarrow \Delta|_{max} = -144$, $\Delta|_{min} = -3600$

$\Rightarrow$ Difference = 3456.

89. 00000.50

Sol. Since square of sides are given. cosine rule will be applicable

$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \Rightarrow b^4 + c^4 + a^4 + 2b^2c^2 - 2a^2b^2 - 2a^2c^2 = 4b^2c^2\cos^2 A$

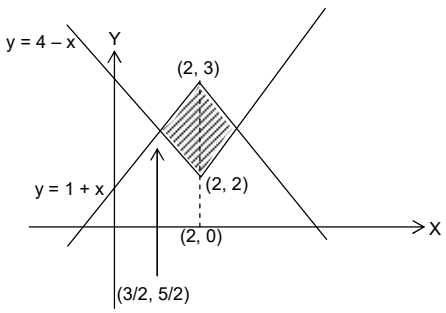
$\Rightarrow a^4 + b^4 + c^4 = 2a^2c^2 + 2a^2b^2 + b^2c^2(-2 + 4\cos^2 A)$

compare with given values gives $-2 + 4\cos^2 A = 1$

$\Rightarrow \cos^2 A = \frac{3}{4} \Rightarrow \sin^2 A = \frac{1}{4} \quad \Rightarrow \sin A = \frac{1}{2}$

90. 00000.50

Sol. The two curves are $y = |x - 2| + 2$ & $y = 3 - |x - 2|$



The area bounded

=shaded area

$$= 2\int_{3/2}^{2} (1 + x - 4 + x)\,dx$$

$$= 2\left[x^2 - 3x\right]_{3/2}^{2} = \frac{1}{2}.$$